

From Fingerspelling to Sign Language Machine Translation

A Decade of NLP and HCI Research on Myanmar Sign Language

Ye Kyaw Thu

Lab Leader, Language Understanding Lab, Myanmar
Visiting Professor, NECTEC, Thailand

<https://sites.google.com/site/yekyawthunlp/> <https://github.com/ye-kyaw-thu>

15 July 2026

Vietnam–Korea University of Information and Communication Technology (VKU)
The University of Danang, Vietnam

Outline

- 1 Introduction
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Why Myanmar Sign Language?

- About 1.3% of Myanmar's population is Deaf or hard-of-hearing (2014 census)
- Estimated 673,126 Deaf people rely on Myanmar Sign Language (MSL)
- Only $\sim 0.006\%$ of Deaf people have a university education
- Very few sign-language interpreters and assistive communication systems
- Deaf and hearing people face a real, everyday communication gap

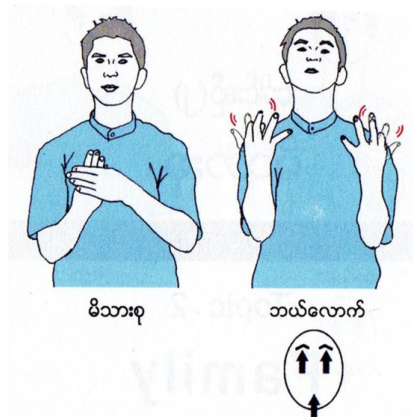
- Goal: reduce the communication barrier between Deaf and hearing communities in Myanmar
- Two complementary research directions pursued over more than a decade:
 1. **HCI**: text input for fingerspelling and Sutton SignWriting
 2. **NLP**: machine translation between MSL, Myanmar SignWriting (MSW) and Myanmar written text (MWT)
- Close collaboration with Deaf schools: Mary Chapman School (Yangon), School for the Deaf (Mandalay, Tarmwe)

A Decade of Work: Roadmap



Myanmar Sign Language (MSL)

- A full, natural language – *not* a signed version of Myanmar
- Two components:
 - **Manual features (MFs)**: hand shape, location, orientation
 - **Non-manual features (NMFs)**: eyebrows, head tilt, mouth, shoulders
- Grammar, word order and vocabulary differ from Myanmar written text
- Regional dialect variation: Yangon (Mary Chapman) vs. Mandalay

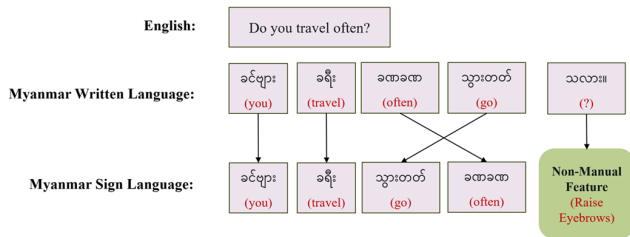


MSL vs. Myanmar Written Text: Word Order

Example (from Moe et al., 2023, MT Summit)

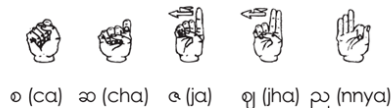
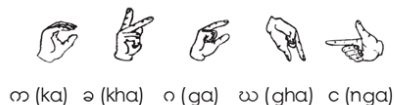
English: *Do you travel often?*

- MWT order: you – travel – often – go – (question mark)
- MSL order: you – travel – go – often – (raised-eyebrow NMF for question)
- Sign order differs; the question mark is replaced by a non-manual cue



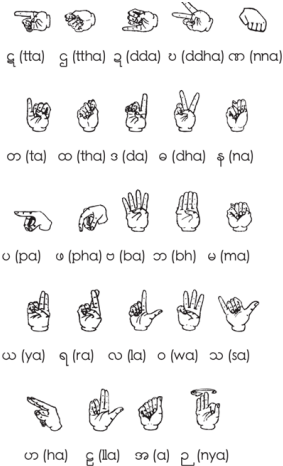
Myanmar Fingerspelling

- Used to spell names of people, places and words with no dedicated sign
- Two regional character sets: Mary Chapman (Yangon, southern) vs. Mandalay (northern)
- Similar consonants, different vowels/medials/symbols
- 33 consonant characters; 31 are static, 2 are dynamic gestures

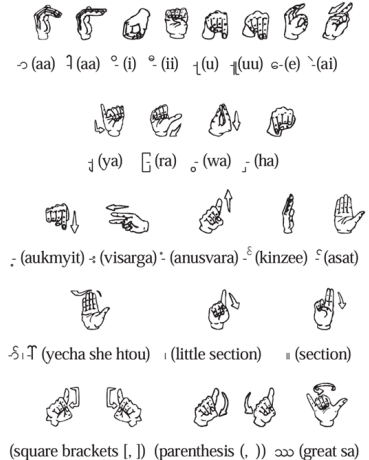


Myanmar Fingerspelling

Consonants



Vowels, Consonant Signs, Various Signs, and Symbols



Independent Vowels and Symbols



ဒ် (i)



ဤ (ii)



ဥ (u)



ဧ (uu)



ဩ (o)



၏ (i)



ဤ (yae)



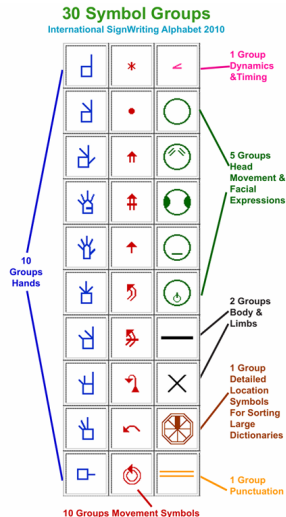
ဤ (nhaik)



ငါး (le gaung)

What Is Sutton SignWriting?

- Developed by Valerie Sutton in 1974; a true *writing system* for signed languages
- Iconic symbols represent hand shape, movement, location, facial expression
- International SignWriting Alphabet (ISWA): 7 categories, 30 symbol groups
- Included in Unicode (U+1D800–U+1DAAF)
- References:
 - <https://www.signwriting.org/>
 - <https://en.wikipedia.org/wiki/SignWriting>

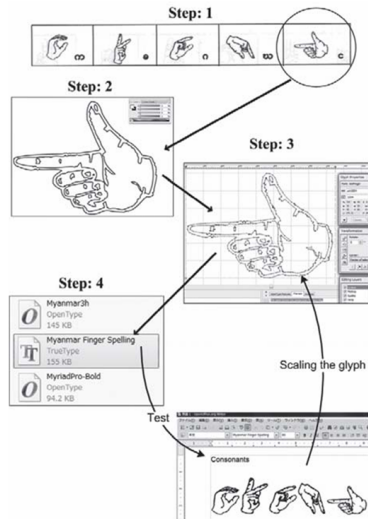


Font Development and Keyboard Layouts for Myanmar Fingerspelling and SignWriting

- No fingerspelling font or standard keyboard existed for Myanmar (2009)
- Sign-language textbooks still relied on hand-drawn pictures
- Goal: enable digital typing of fingerspelling and SignWriting text

mmFingerspelling.ttf: Font Development

- Scanned the standardized (2007) Myanmar fingerspelling character set
- Traced characters to vector format (Adobe Illustrator)
- Assigned Unicode code points and glyph paths (FontLab Studio)
- Result: mmFingerspelling.ttf – first Myanmar fingerspelling font



Direct Keyboard Mapping (DKM)

- Idea: map fingerspelling characters onto the existing **Myanmar3** Unicode keyboard layout
- Rationale: short learning curve for users already familiar with Myanmar typing
- Each fingerspelling character needs only *one* glyph (no positional variants)
- Two typing modes: physical QWERTY keyboard, or on-screen software keyboard



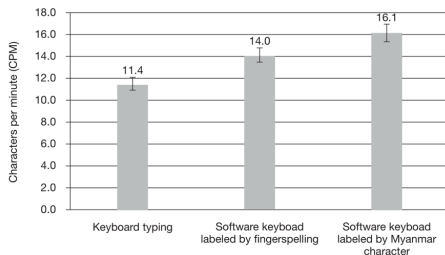
- 48 participants: 18 hearing-impaired, 16 general users, 14 DTP staff
- 3 text-input conditions: physical keyboard, software keyboard (fingerspelling labels), software keyboard (Myanmar-character labels)
- Task: type 5 fingerspelling names, 10 repetitions
- Metrics: Characters Per Minute (CPM) and 4-point Likert scales

Five fingerspelling names for user study (Equivalent Myanmar characters are presented with brackets)



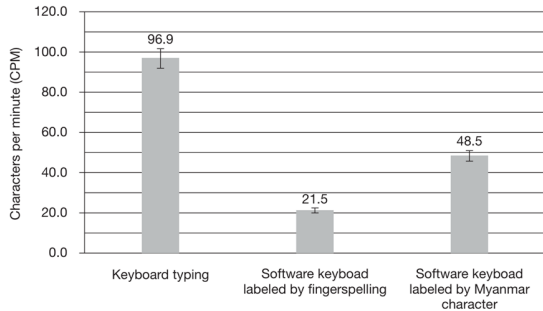
DKM Results

- Hearing-impaired users typed fastest with the *Myanmar-character-labeled* software keyboard (16.1 CPM)
- General users and DTP staff were fastest with the physical keyboard, as expected
- Likert scores show strong acceptance of DKM across all user groups
- Conclusion: Myanmar-character labeling gives a better first-time user interface than fingerspelling labeling

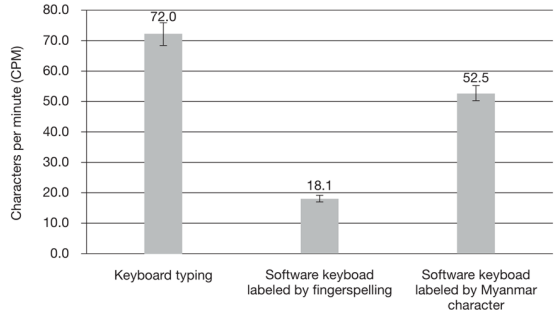


Average CPM of hearing-impaired users.

DKM Results



General users



DTP staff

From Fingerspelling to SignWriting Input

- DKM solved fingerspelling *character* input
- But Sutton SignWriting symbols need *multiple* keystrokes: symbol + fill + rotation modifiers
- No existing keyboard or editor for Myanmar SignWriting $\Rightarrow$ next research question

Two Fingerspelling Keyboard Layouts for MSW (ICCA 2018)

- **Phonetic-based layout:** maps Myanmar characters by phonetic similarity to English keys (kKg concept)
- **Symbol-based layout:** groups keys by visual shape-similarity of SignWriting symbols
- Implemented with X Keyboard Extension (XKB) for Linux/Unix



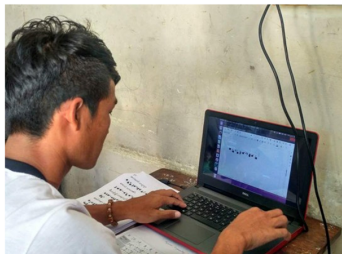
Two Fingerspelling Keyboard Layouts for MSW (ICCA 2018)

Symbol-based Keyboard Layout for Myanmar SignWriting



SignWriting Keyboard Layouts User Study

- 19 participants: 8 hearing-impaired + 11 hearing users
- Each typed 3 fingerspelling poems (parallel MSW/Myanmar text), 10 times
- Measured CPM and 4-point Likert scale (difficult–easy, slow–fast, dislike–like, impossible–possible)



SignWriting Keyboard Layouts User Study

An example of fingerspelling poem

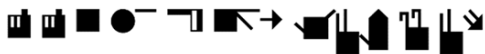
လယ်သမားလူငယ်



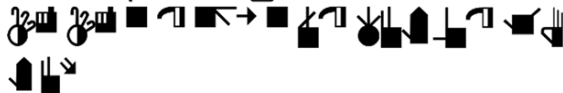
စပါးနယ်နေသည်။



မမစပါးသယ်၏။



မေမေစားစရာဝယ်လာသည်။

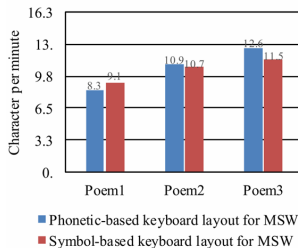


ဖရဲသီးပါသလား။



Results for SignWriting Keyboards

- Phonetic-based layout was consistently faster (e.g., hearing users: 26.5 vs. 23.5 CPM on poem 3)
- 63–82% of participants preferred the phonetic-based layout for real-time use
- Symbol-based layout: praised for logical grouping, but harder to memorize



Improving Efficiency: the kKg SignWriting Keyboard (2020)

- Goal: reduce keystrokes-per-character further than the phonetic-based layout
- Applies the **kKg keyboard mapping concept** (Myanmar consonant-phonetic mapping) directly to SignWriting symbols
- Implemented with the Keyman Developer engine (cross-platform)
- Example: typing "*nga*" needs 1 key here vs. 3 keys (symbol+fill+rotation) in the earlier phonetic layout

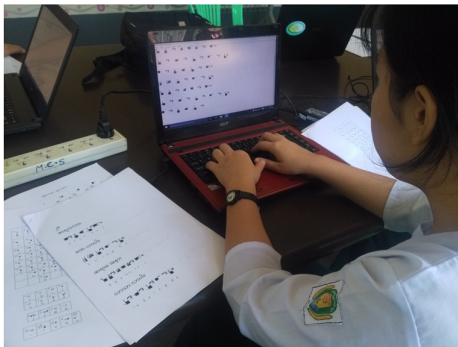


Improving Efficiency: the kKg SignWriting Keyboard (2020)

Myanmar SignWriting keyboard layout for Myanmar fingerspelling (shift mode)

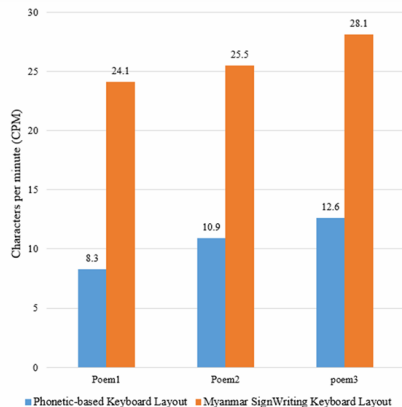


Experimental environment with hearing-impaired users



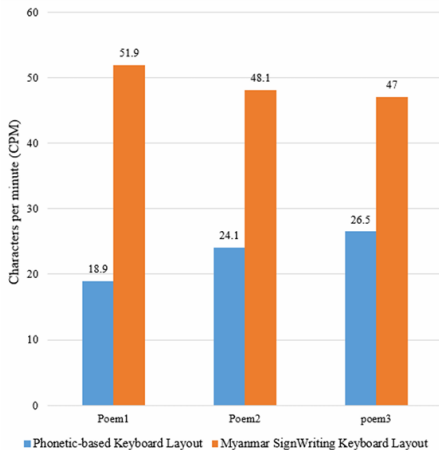
kKg SignWriting Keyboard: Results

- Total keystrokes cut roughly in half vs. the earlier phonetic-based layout (e.g., poem 2: 169 → 68 keystrokes)
- Hearing-impaired users: CPM roughly doubled (8.3–12.6 → 24.1–28.1)
- Hearing users: CPM nearly doubled (18.9–26.5 → 47–51.9)
- Likert scores confirm higher perceived speed and ease of use



kKg SignWriting Keyboard: Results

Average CPM value of hearing users



Fingerspelling / SignWriting Text Input: Summary

- 2010: first Myanmar fingerspelling font + DKM keyboard
- 2018: first SignWriting keyboards for Myanmar fingerspelling (2 layouts)
- 2020: redesigned SignWriting keyboard $\Rightarrow$ big efficiency gains
- Together: a complete, evolving HCI toolchain for MSL fingerspelling text entry

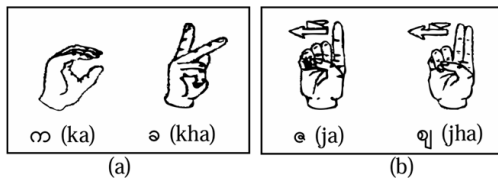
Recognizing Myanmar Fingerspelling from Images

Motivation

- Typing solves *text* input, but many use cases need *recognition from images/video*
- Very little prior work on Myanmar fingerspelling recognition
- Goal: classify static fingerspelling hand-shape images automatically

Data and Features

- 31 static consonant characters, Mary Chapman School character set
- 11 Deaf signers, ages 12–25; 24,713 images (16,557 train / 8,156 test)
- Three classical feature extractors:
 - SIFT (128-d), SURF (64-d), BRIEF (2,048-d)



Classifiers and Setup

- 7 classical ML classifiers compared: MLP, Gaussian Naïve Bayes, Decision Tree, Logistic Regression, Random Forest, SVM, KNN
- 10-fold cross-validation for robustness
- Accuracy = correct predictions / test samples

Classifier	SIFT	SURF	BRIEF
Random Forest	95%	97%	81%
KNN	94%	96%	81%
Decision Tree	89%	92%	78%
SVM	90%	14%	80%

Table 1: *

Accuracy on 31 Myanmar fingerspelling classes

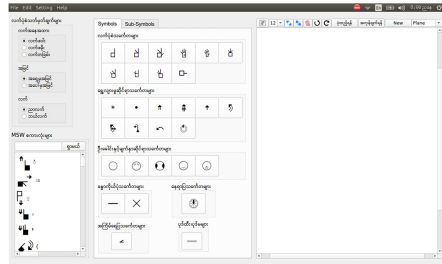
- Random Forest + SURF: best overall (97% accuracy)
- MLP and Gaussian Naïve Bayes underperformed across all features

Discussion and Future Work

- No special hardware needed – an ordinary webcam suffices
- Only lightweight preprocessing (resize + grayscale) required
- Future: real-time recognition, dynamic gestures, deep neural networks

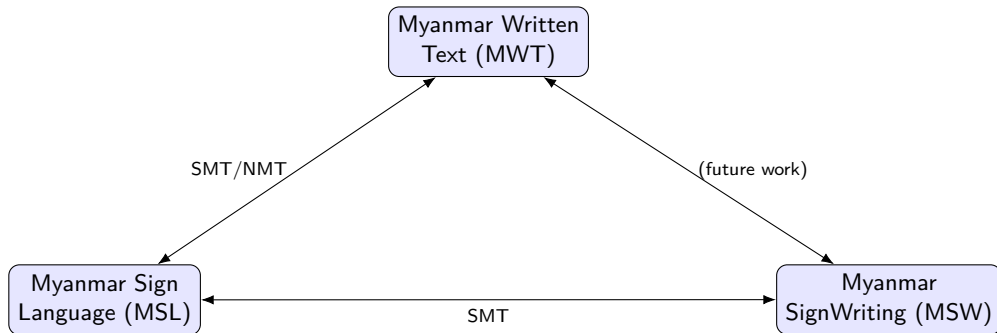
SignWriting GUI Prototype (Unpublished Work)

- A graphical editor for composing Myanmar SignWriting, developed as part of Hlaing Myat Nwe's master's thesis
- Complements the keyboard-layout research: a friendlier, canvas-based alternative to raw keystrokes
- Not published as a paper – shared openly via GitHub
- <https://github.com/ye-kyaw-thu/SignWriting-GUI-Prototype>



Translating Between MSL, SignWriting and Myanmar Written Text

The Translation Triangle



Building an MSL Corpus (2016–present)

- No parallel MSL corpus existed in Myanmar $\Rightarrow$ had to build one from scratch
- 22–30 sign-language trainers and Deaf people (ages 11–48), multiple schools
- Covers Yangon and Mandalay dialects
- Contains MSL video + textual MSL gloss + translated Myanmar written text
- Emergency-domain subset released publicly:
<https://github.com/ye-kyaw-thu/MSL4Emergency>

SMT between MSL and MSW (ASEAN-AI 2018)

- First evaluation of automatic translation between MSL and Myanmar SignWriting
- Methods: Phrase-based SMT (PBSMT), Hierarchical PBSMT, Operation Sequence Model (OSM)
- Corpus: 1,448 general-domain parallel sentences (1,000 / 170 / 278 train-dev-test)
- Two segmentation schemes compared: word vs. syllable segmentation for MSL

SMT $MSL \leftrightarrow MSW$: Results

Direction	PBSMT	HPBSMT	OSM
$MSL \rightarrow MSW$ (word)	34.44	34.99	37.54
$MSW \rightarrow MSL$ (word)	52.66	52.79	49.99

Table 2: *

BLEU scores, word-segmented MSL

- Word segmentation outperformed syllable segmentation for this pair
- OSM best for $MSL \rightarrow MSW$; HPBSMT best for $MSW \rightarrow MSL$

SMT Error Analysis

- Main confusion sources: (1) genuine sign-language synonymy, (2) reference/annotation noise, (3) Yangon–Mandalay dialect variation, (4) limited training data
- Example: “*mahotu*” (*no*), “*mahlutne*” (*don’t do it!*), “*mahpyit*” (*do not*) all map to one MSL sign
⇒ hard for the model to disambiguate

NMT between MSL and MWT (ONA 2018)

- Corpus: 2,510 parallel MSL–MWT sentences, everyday conversation domain
- Compared RNN, CNN (ConvSeq2Seq) and Transformer (Sockeye toolkit)
- Four segmentation schemes: word, syllable, SentencePiece, BPE

NMT MSL $\leftrightarrow$ MWT: Results

Segmentation	MWT $\rightarrow$ MSL (best)	MSL $\rightarrow$ MWT (best)
Word	28.80 (CNN)	29.38 (Transformer)
Syllable	32.76 (CNN)	38.21 (Transformer)
SentencePiece	22.69 (CNN)	23.64 (CNN)
BPE	29.39 (Transformer)	32.92 (Transformer)

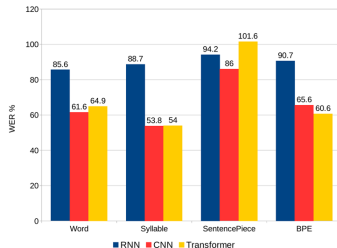
Table 3: *

Highest BLEU per segmentation scheme

- Syllable segmentation gave the best results overall; Transformer > CNN > RNN

NMT Error Analysis (WER)

- Word Error Rate (WER) confirms syllable segmentation is most robust
- Deletion errors dominate $\Rightarrow$ model struggles most with MSL/MWT word-order differences
- Substitution errors often *acceptable* (near-synonym word choice, e.g., formal vs. informal “you”)



Unsupervised NMT for MSL–MWL (Journal, 2020)

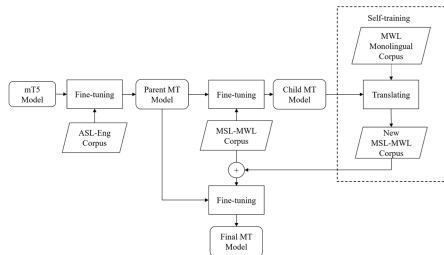
- Motivation: parallel data is scarce and expensive to collect
- U-NMT (Lample et al., 2018): BPE-based shared vocabulary, denoising autoencoding, iterative back-translation
- Only Myanmar-language *monolingual* data enriched: myPOS corpus and translated primary-school text
- No additional MSL monolingual data was available for this experiment

U-NMT: Training Data and Results

- Four training-data configurations tested (with/without extra monolingual Myanmar text)
- Adding the myPOS corpus gave the highest scores: BLEU 30.13 for MWL→MSL
- U-NMT clearly outperformed supervised NMT for MWL→MSL, given the same tiny parallel set
- MSL→MWL direction: still below supervised NMT quality

Extending the Idea: Transfer Learning + Self-Training (MT Summit 2023)

- Joint work with JAIST, Thammasat University and NECTEC
- Base model: multilingual mT5 (580M parameters)
- **Transfer learning**: pretrain a parent model on a large ASL–English corpus (ASLG-PC12), then fine-tune on the small MSL–MWL set
- **Self-training**: use the fine-tuned model to translate a large MWL monolingual corpus (myPOS) into MSL, filtered by model confidence, to enlarge the training set



Transfer + Self-Training: Results

Model	MSL→MWL (syllable)	MWL→MSL (syllable)
mT5 (baseline)	50.67	51.23
mT5+Transfer	51.29	56.29
mT5+Self-train	52.26	55.93
mT5+Transfer+Self-train	56.60	57.11

Table 4: *

BLEU scores

- Combining both techniques gives the best result in every configuration
- ~6 BLEU points above the mT5-only baseline

Putting Results in Context

Method	MSL→MWL	MWL→MSL
mT5+Transfer+Self-train (2023)	37.83	39.97
Supervised SMT (2018)	34.78	35.11
Supervised NMT (2018)	38.21	32.92
Unsupervised NMT (2020)	10.47	29.53

Table 5: *

BLEU, syllable-level comparison across the whole research line (different test sets – not perfectly fair, but indicative)

- Nine years of iteration: from first SMT baselines to transfer-learned mT5

Overall Summary

- **HCI line:** font → DKM keyboard → 2 SignWriting keyboards → optimized kKg keyboard
- **Recognition:** classical ML fingerspelling image classification (97% accuracy)
- **NLP line:** SMT/NMT/unsupervised-NMT/transfer-learning MT across MSL, MSW, MWT
- Common thread: build resources first (corpora, fonts, keyboards), then models
- All work done in close partnership with Deaf schools and students

Future Directions

- Incorporate non-manual features (facial expression, head movement) into MT
- Larger MSL–MWT–MSW parallel corpora; more domains
- Video-based MSL $\leftrightarrow$ text translation (skip textual gloss as an intermediate step)
- Real-time fingerspelling recognition with deep neural networks
- A full Myanmar SignWriting text editor for Deaf education

Acknowledgments

- Students and co-authors: Hlaing Myat Nwe, Swe Zin Moe, Hnin Wai Wai Hlaing, Ni Htwe Aung, Hnin Aye Thant, Nandar Win Min, Sai Aung Win Maung, and others
- Collaborating institutions: Waseda University, NECTEC, JAIST, Thammasat University, University of Technology (Yatanarpon Cyber City)
- Mary Chapman School for the Deaf (Yangon); Schools for the Deaf (Mandalay, Tarmwe)
- All the Deaf participants and sign-language trainers who made this research possible

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Thank You

Questions & Discussion

Ye Kyaw Thu

<https://sites.google.com/site/yekyawthunlp/> <https://github.com/ye-kyaw-thu>